PRESS-FIT CONNECTOR
& FIXTURE DESIGN

Complete Guide to Assembly Machines, Tooling & Fixture Design for PCBA

Topics Covered:

- Press-Fit Technology Overview
- Connector Types: Pin Headers, Power Elements, Backplane
- Assembly Machines: Manual, Semi-Auto, Fully Automatic
- Fixture Design: Top Tools, Bottom Fixtures, Support Plates
- Force-Distance Monitoring & Quality Control
- Common Challenges & Best Practices

1. PRESS-FIT TECHNOLOGY OVERVIEW

Press-fit technology is a solderless interconnection method where compliant pins or solid pins are mechanically pressed into plated through-holes (PTHs) on a PCB. The pin deforms (or the PCB hole deforms) to create a **gas-tight, high-conductivity electrical connection** without any thermal process. This eliminates soldering defects like cold joints, voids, and solder cracks.

- **No Thermal Stress:** No heating of PCB or components — ideal for heat-sensitive assemblies
- **High Reliability:** Gas-tight contact with contact resistance as low as 0.1–0.2 mΩ
- **Vibration Resistance:** Superior performance in automotive, aerospace, and industrial environments
- **Reworkable:** Pins can typically be removed and replaced 2–3 times without damaging the PCB
- **High Current Capacity:** Power elements handle 50A–500A+ with multiple parallel pins
- **Single-Pass Assembly:** SMT components are reflowed first, then press-fit connectors inserted in a separate mechanical step

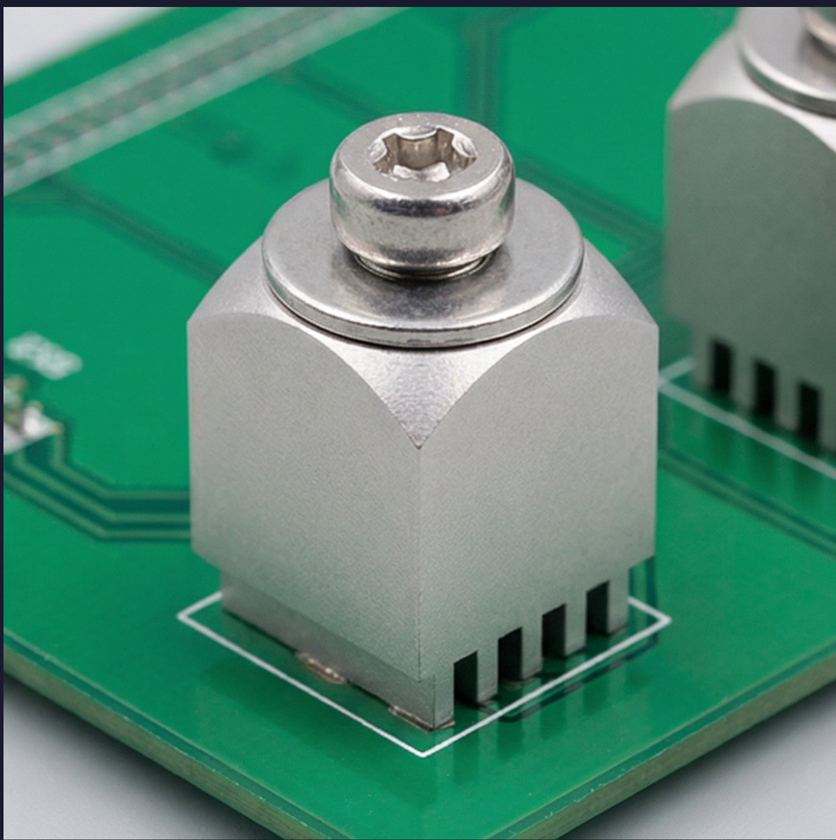
The three critical elements of press-fit assembly are: **(1) the press-fit connector** with compliant pins, **(2) the press-fit machine** with force-distance monitoring, and **(3) the fixture/tooling** that aligns and supports the PCB during insertion. All three must be properly designed and coordinated for reliable production.

2. TYPES OF PRESS-FIT CONNECTORS

Press-fit connectors come in various configurations depending on the application, pin geometry, and mounting orientation.

Connector Type	Description	Typical Use	Pin Type
Pin Header (0° / 90° / Angled)	Plastic housing with array of press-fit pins. Mating di[illegible]	Board-to-board, [illegible]	[illegible] Eye of Needle, Action Pin
Power Element (M3–M10)	Heavy-duty metal block with threaded top and multi[illegible]	Power distribution, [illegible]	[illegible] (multiple pins in pa
Backplane Connector	High-density multi-row connectors for high-speed si[illegible]	Telecom switches, [illegible]	[illegible] (fine-pitch, controlled impe
Single Pin (Standalone)	Individual press-fit pins inserted one at a time. Used [illegible]	Repair, [illegible]	[illegible] or Solid
Integrated Housing (Module)	Connector, coil, capacitor, or sensor housed in a si[illegible]	[illegible]	[illegible] (molded into housing)
Board-to-Board Interposer	Connector that bridges two PCBs with press-fit pins [illegible]	[illegible]	[illegible] (double-ended)
IDC / FFC Connector	Flat cable connector with press-fit tails for ribbon ca[illegible]	[illegible] wiring, display cables, control panels	Compliant (stitched into housing)

Image Reference — Power Tap Press-Fit Element on PCB:



Power tap press-fit element with threaded top and compliant pins mounted on PCB for high-current applications

Image Reference — Press-Fit Terminal with Compliant Pins:

Press-fit PCB power terminals with multiple compliant pins for solderless high-current connection

3. COMPLIANT PIN GEOMETRY TYPES

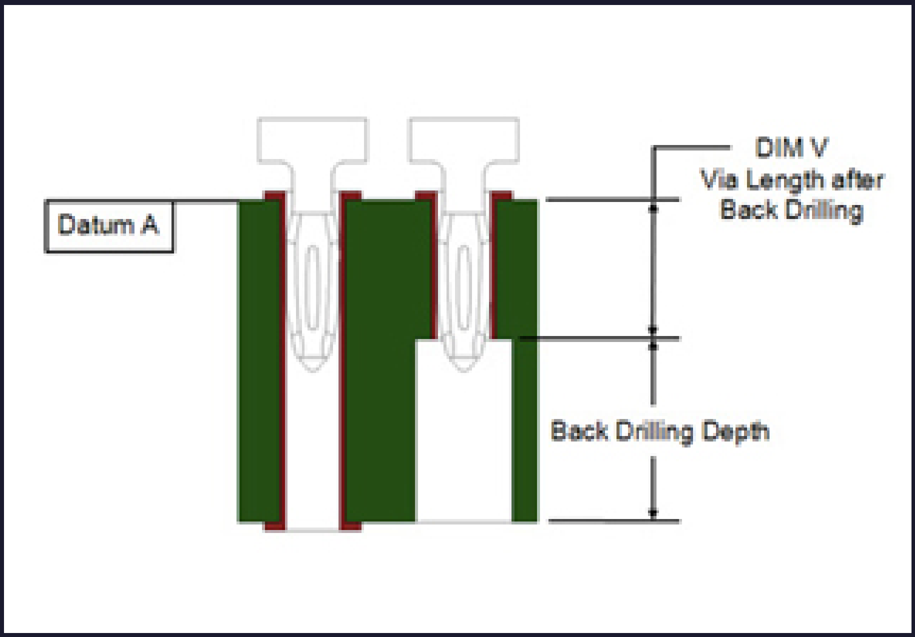
The compliant zone of a press-fit pin is designed to be slightly larger than the PTH diameter. During insertion, the pin deflects inward, generating high normal force against the barrel walls. Different geometries suit different applications.

Pin Type	Geometry	Insertion Force	Best For
Eye of the Needle (EON)	Two opposing spring arms with oval eye-shaped slot. Arms deflect inward during insertion.	60–120N per pin	General-purpose, automotive, high-density connectors
Action Pin	Cross-shaped or X-shaped compliant zone. Four contact points provide redundant contact.	80–150N per pin	High-vibration environments, heavy contamination
C-Press	C-shaped split shank that compresses during insertion. Simple, robust design.	50–100N per pin	Cost-sensitive applications, large pins
Tcom	Hourglass or bow-tie shaped compliant zone. Balanced elastic-plastic deformation.	70–130N per pin	High-reliability telecom and industrial
Solid Pin (Massive)	No compliant zone — solid pin pressed into smaller hole. PCB hole deforms to match pin.	150–250N per pin	High-current power elements, coarse pitch
W-Shaped (Multi-Beam)	Multiple thin beams arranged in W-pattern. Very low insertion force, high contact redundancy.	40–80N per pin	Fine-pitch backplane, high-speed signals

Critical PTH Parameters for Compliant Pins:

Pin Size	Finished Hole Diameter	Copper Wall Thickness	Surface Finish
0.4mm EON	0.60 ± 0.05mm	25–35µm	Clean, uniform tin or ENIG
0.64mm EON	1.05 ± 0.05mm	25–40µm	Clean, uniform tin or ENIG
0.81mm EON	1.50 ± 0.05mm	30–50µm	Clean, uniform tin or ENIG
M3 Power Element	1.0 ± 0.08mm	25–40µm	Tin-plated (15µm max)
M4 Power Element	1.2 ± 0.10mm	30–50µm	Tin-plated (15µm max)
M6 Power Element	1.6 ± 0.13mm	35–50µm	Tin-plated (15µm max)

Image Reference — Compliant Pin Cross-Section (Back-Drilling):



Compliant pin cross-section showing Eye of the Needle geometry with back-drilling depth and via length dimensions

4. PRESS-FIT ASSEMBLY MACHINES

Press-fit machines range from simple manual arbor presses to fully automatic servo-electric systems with force-distance monitoring and vision systems. The choice depends on production volume, connector complexity, and quality requirements.

4.1 Manual & Semi-Automatic Presses

Machine	Type	Force	Best For
Manual Arbor Press (P10)	Hand-operated ram press	Up to 10kN	Repair stations, prototyping, low-volume sample production
P50 Bench Machine	Manual PCB positioning, automatic pin insertion	Pneumatic, up to 5kN	Low-volume production, repair, sample builds
P100 Insertion Machine	Manual load, programmable PCB positioning, auto insertion	Pneumatic, up to 10kN	Medium-volume, multiple pin types
Hand Press with Force Gauge	Simple manual press with digital force readout	Operator-dependent	R&D, qualification testing, small batches

4.2 Fully Automatic Servo-Electric Presses

Servo-electric presses provide precise control of force, distance, and speed with real-time monitoring. They are the standard for automotive and high-reliability production.

Machine	Force	Features	Throughput
CAP-6T / CAPI-6T	Up to 27kN (3 tons)	PC control, force-distance monitoring, pin penetration sensing (PPS), automatic shuttle	6–10 sec per cycle
CSP-3T Shuttle Press	Up to 27kN (3 tons)	Automatic shuttle, XY positioning, PPS tooling, SMEMA compatible	Up to 120 pins/min
CSM-5T Press	Up to 44kN (5 tons)	Semi-auto, right-angle connector loading, force feedback, traceability	Medium-high volume
P300 / P350	Up to 44kN (5 tons)	Fully automatic single-pin insertion, vision system, tool changer, SMEMA line	3.5–5 pins/sec
CT-2030-2 (Inline)	Up to 50kN	Inline operation, vision mark positioning, 2D code scanning, MES integration	High-volume continuous
Automated Press (Robotic)	Up to 115kN	Robotic loading, automatic tool change, full traceability, pin presence detection	Fully automated line

4.3 Machine Selection Criteria

Choose the press-fit machine based on these factors:

- **Force Capacity:** Calculate total insertion force = number of pins × force per pin. Add 20% safety margin. A 200-pin connector at 100N/pin needs 24kN + 20% = 28.8kN minimum.
- **Control Type:** Servo-electric is mandatory for force-distance monitoring. Pneumatic/hydraulic lacks precision and cannot detect bent pins or misalignment.
- **Shuttle System:** Automatic shuttle allows operator to prepare next PCB while current one is pressing. Reduces cycle time by 30–40%.
- **Vision System:** Mark point positioning and connector true-position detection prevent misalignment. Essential for fine-pitch backplane connectors.
- **PPS (Pin Penetration Sensing):** Verifies every pin penetrates the PCB by a predetermined distance. Detects bent, missing, or improperly seated pins in real-time.

- **Traceability:** Force-distance curves stored per board for quality records. Required for automotive (IATF 16949) and medical (ISO 13485) applications.

5. FIXTURE & TOOLING DESIGN

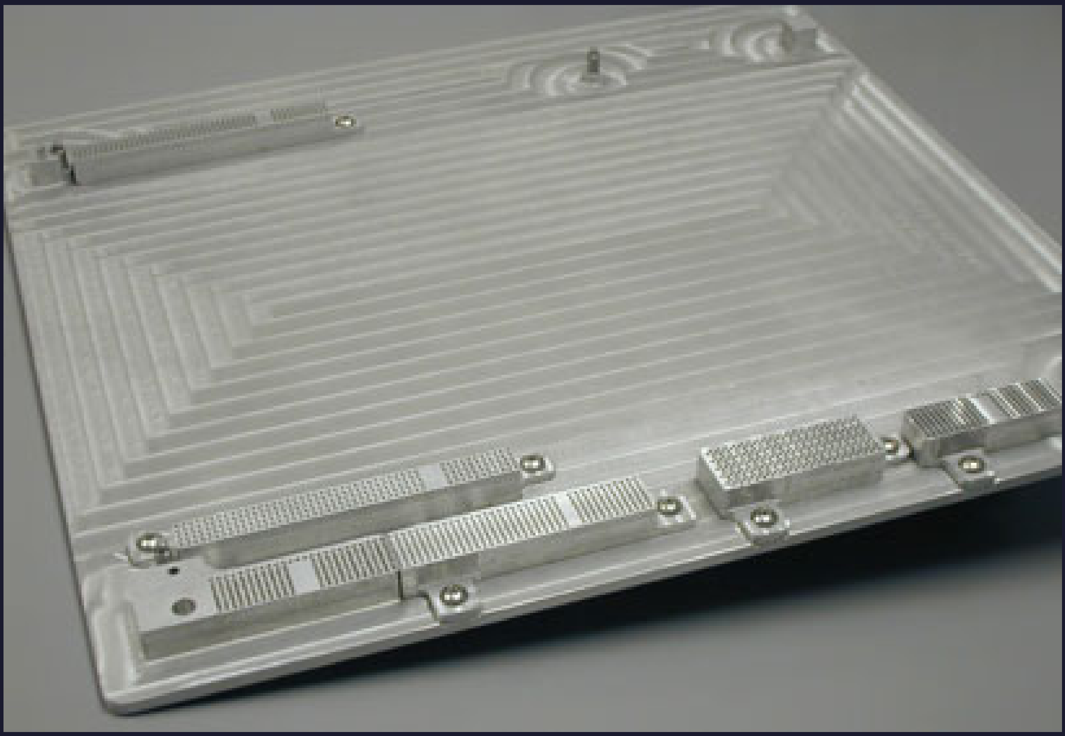
Proper fixture design is critical to prevent PCB damage, ensure pin alignment, and distribute insertion forces evenly. Fixtures consist of **top tools** (insertion tools) and **bottom fixtures** (support plates).

5.1 Bottom Support Fixture (PCB Support Plate)

The bottom fixture supports the PCB during pressing and prevents board flexure that could crack traces or damage SMT components.

- **Material:** Precision-ground aluminum (most common), stainless steel (heavy-duty), PEEK (insulated), or composite (lightweight). Aluminum is preferred for ESD safety and machinability.
- **Clearance Holes:** Must provide clearance for press-fit pins that protrude through the PCB. Pin tips must not contact the fixture — typically 2–3mm clearance below PCB.
- **Support Pillars:** Raised platforms support the PCB around each connector site while avoiding bottom-side SMT components. Height must account for tallest component on bottom side.
- **Guide Pins:** Tooling pins in the fixture pre-align PCB holes with the fixture. These must be longer than press-fit pins and contact the board first to ensure registration.
- **Fixture Identification:** Barcode or RFID tag on both sides for traceability. Prevents wrong fixture being loaded.
- **Open Areas:** Frame design should have open areas for easy PCB insertion and removal. Remove excess material to reduce weight.
- **Smooth Edges:** All edges chamfered or radiused to prevent operator injury and PCB damage.

Image Reference — Press-Fit Support Plate (Bottom Fixture):



Precision aluminum press-fit support plate with support pillars and clearance areas for bottom-side SMT components

5.2 Top Insertion Tool (Press Tool)

The top tool presses the connector into the PCB. It must distribute force evenly across all pins without damaging the connector housing.

• **Force Application:** Most connectors are pressed on the **shoulders of the press-fit pins**. Some designs press on the plastic housing material — the housing must be strong enough.

• **Flat-Rock Pressing:** For right-angle daughter card connectors, a simple flat surface tool can be used. The connector housing must be sufficiently rigid.

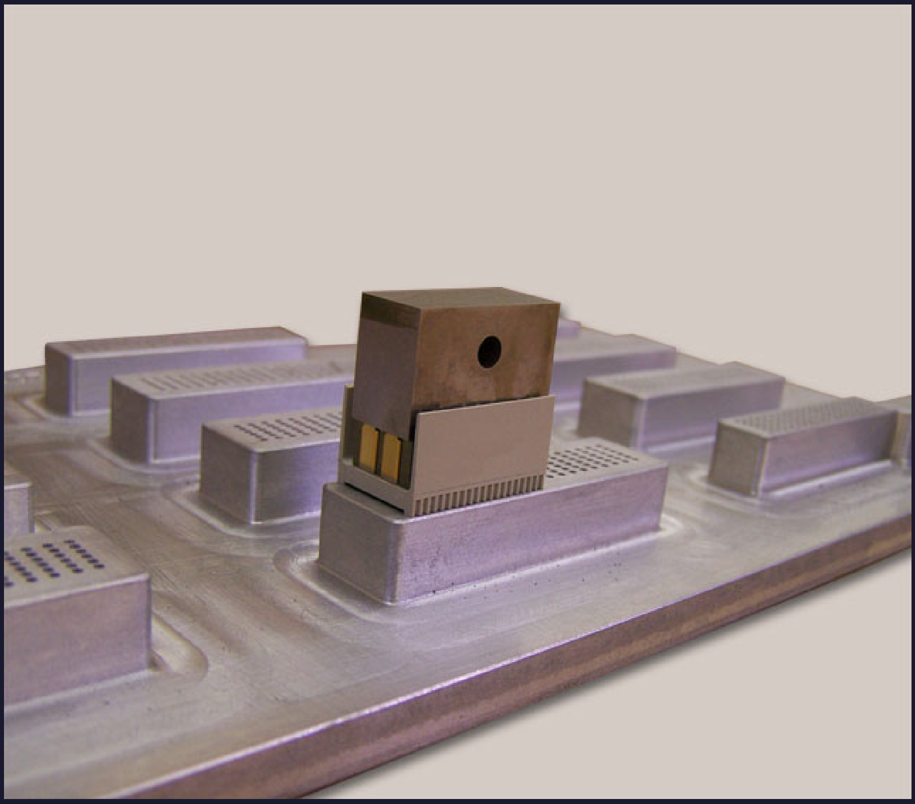
• **Individual Pin Pressing:** Slotted tools or tools with individual pin holes press each pin separately. Prevents pin-to-pin force transfer and reduces housing stress.

• **Material:** A2 tool steel, heat-treated to Rockwell 50–55 HRC. Provides durability for high-volume production.

• **Maximum Tolerance:** Base-to-press bar tolerance ≤3 mils (0.076mm) to ensure even force distribution across all pins.

• **PPS Integration:** Pin Penetration Sensing tools have individual probes that verify each pin's penetration depth during pressing.

Image Reference — Press-Fit Tool with Connector:



Press-fit insertion tool mounted on support plate, showing connector alignment and top-tool engagement

5.3 Fixture Design Checklist

■ **Barcode/RFID:** Fixture identified on both sides for traceability

■ **Standard Depth:** Consistent fixture thickness across all fixtures in the plant

■ **Guide Pins:** Pre-align PCB before pins engage. Longer than longest pin.

■ **Bottom Clearance:** 2–3mm gap below PCB for pin protrusion. Account for tallest bottom-side SMT.

■ **Support Pillars:** Support PCB directly under connector sites. Prevent board flexure.

■ **Weight Reduction:** Remove excess material. Lighter fixtures = faster handling, less operator fatigue.

■ **ESD Compliance:** Aluminum or stainless steel preferred. Avoid materials that generate static.

■ **Smooth Edges:** Chamfer all edges. Prevent PCB snagging and operator injury.

- **Tool Storage:** Stored on soft material (foam, rubber) to prevent damage. Upright racks preferred.
- **Maintenance Schedule:** Regular inspection for wear, damage, and dimensional drift.

6. FORCE-DISTANCE MONITORING & QUALITY CONTROL

Force-distance monitoring is the primary quality assurance method for press-fit assembly. Every press cycle generates a unique curve that reveals the health of the connection.

6.1 The Force-Distance Curve

A typical press-fit force-distance curve has these phases:

- **Phase 1 — Approach:** Tool descends to contact connector. Minimal force (0–50N).
- **Phase 2 — Initial Contact:** Tool touches connector housing or pin shoulders. Force rises rapidly.
- **Phase 3 — Pin Insertion:** Compliant pins enter PTHs. Force increases as pins compress. Peak force occurs when all pins are simultaneously engaging.
- **Phase 4 — Seating:** Pins reach full depth. Force may drop slightly as compliant zones relax into final position.
- **Phase 5 — Over-Press (if any):** Excessive travel beyond seating point. Should be avoided — indicates wrong PCB thickness or wrong connector.

6.2 Accept/Reject Criteria

The servo-electric press compares each curve against programmed limits:

Parameter	Typical Limit	What It Detects
Maximum Force	±10–15% of nominal	Bent pins, wrong hole size, blocked holes, wrong connector
Minimum Force	±10–15% of nominal	Missing pins, oversized holes, worn tooling, wrong PCB
Total Distance	±0.1–0.2mm of nominal	Wrong PCB thickness, wrong connector, incomplete insertion
Force at Specific Distance	Window check at 50%, 75%, 100% travel	Partial insertion, pin damage, hole plating issues
Curve Shape	Match to master curve (±5% deviation)	Mixed pin types, contaminated holes, inconsistent plating

6.3 Pin Penetration Sensing (PPS)

PPS tooling provides an additional quality layer by physically verifying that every pin has penetrated the PCB by a predetermined minimum distance. It works by:

- **Individual Probes:** Each pin position has a dedicated probe in the top tool.
- **Depth Verification:** Probes measure pin tip protrusion below the PCB surface.
- **Real-Time Detection:** Any pin that fails to reach minimum penetration triggers an immediate stop and alarm.
- **Non-Destructive:** No additional test needed — verification happens during the press cycle itself.
- **Best For:** High-pin-count connectors, automotive safety-critical applications, backplane assemblies.

7. COMMON CHALLENGES & SOLUTIONS

Challenge 1: PCB Barrel Cracking

Excessive insertion force or poor hole quality causes the PCB barrel to crack, compromising electrical connection and board integrity.

Solutions:
• Use compliant pins (EON, Action Pin) instead of solid pins for large arrays
• Verify hole plating thickness ≥25µm copper
• Ensure hole diameter is within spec — oversized holes reduce retention, undersized holes increase stress
• Support PCB directly under connector with fixture pillars
• Use servo-electric press with force limiting to prevent over-press

Challenge 2: Bent or Collapsed Pins

Misalignment between connector and PCB causes pins to bend or collapse instead of entering holes. This is the #1 cause of press-fit defects.

Solutions:
• Use guide pins in fixture that are longer than press-fit pins
• Implement vision system for mark-point positioning and connector true-position detection
• Design connector with polarization features (keying, chamfers)
• Program press to slow down (50% speed) during initial pin engagement
• Use PPS tooling to detect bent pins before full insertion

Challenge 3: Inconsistent Insertion Force

Force varies significantly between boards due to hole size variation, plating inconsistencies, or connector pin geometry variation.

Solutions:
• Specify Cp/Cpk ≥1.33 for hole diameter from PCB fabricator
• Use statistical process control (SPC) on force-distance data
• Qualify multiple PCB fabricators to same hole specification
• Request connector supplier capability data on pin geometry
• Implement automatic tool wear compensation in press software

Challenge 4: SMT Component Damage During Pressing

Insertion force transfers through the PCB and can crack solder joints or displace nearby SMT components.

Solutions:
• Maintain 4mm minimum keep-out zone around press-fit sites
• Place SMT components on opposite side of board from press-fit connectors
• Use support pillars directly under press-fit areas to isolate force
• Apply SMT adhesive (glue) to critical components near press-fit zones
• Sequence assembly: SMT reflow first, then press-fit insertion as separate step

Challenge 5: Wrong Fixture Loaded

Operator error in loading the wrong fixture for a different PCB or connector causes misalignment, pin damage, and scrap.

Solutions:
• Barcode or RFID tag every fixture — scanner validates before press cycle starts
• Tool ID feature on press machine confirms correct upper tool is mounted
• Password-protected program levels (admin, engineer, operator)
• Laser distance sensing to verify fixture and PCB presence
• Standardize fixture dimensions across product families

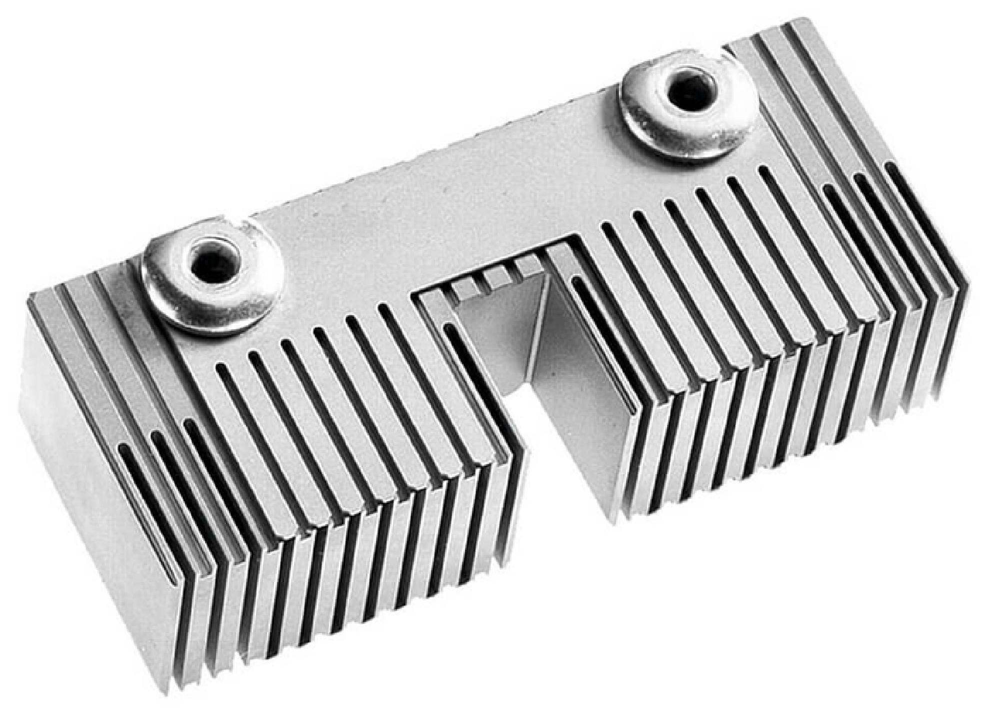
Challenge 6: Connector Housing Deformation

Pressing force applied to the housing instead of pin shoulders can warp or crack plastic connector bodies.

Solutions:
• Design top tool to press on pin shoulders, not housing
• For flat-rock pressing, verify housing material can withstand total force
• Use individual pin pressing tools for high-pin-count connectors
• Specify glass-filled nylon or PBT housings for better rigidity
• Validate housing deformation with FEA simulation before production

8. REFERENCE IMAGES

Press-Fit Tool for Connector (hm2.0 Blade Strip):

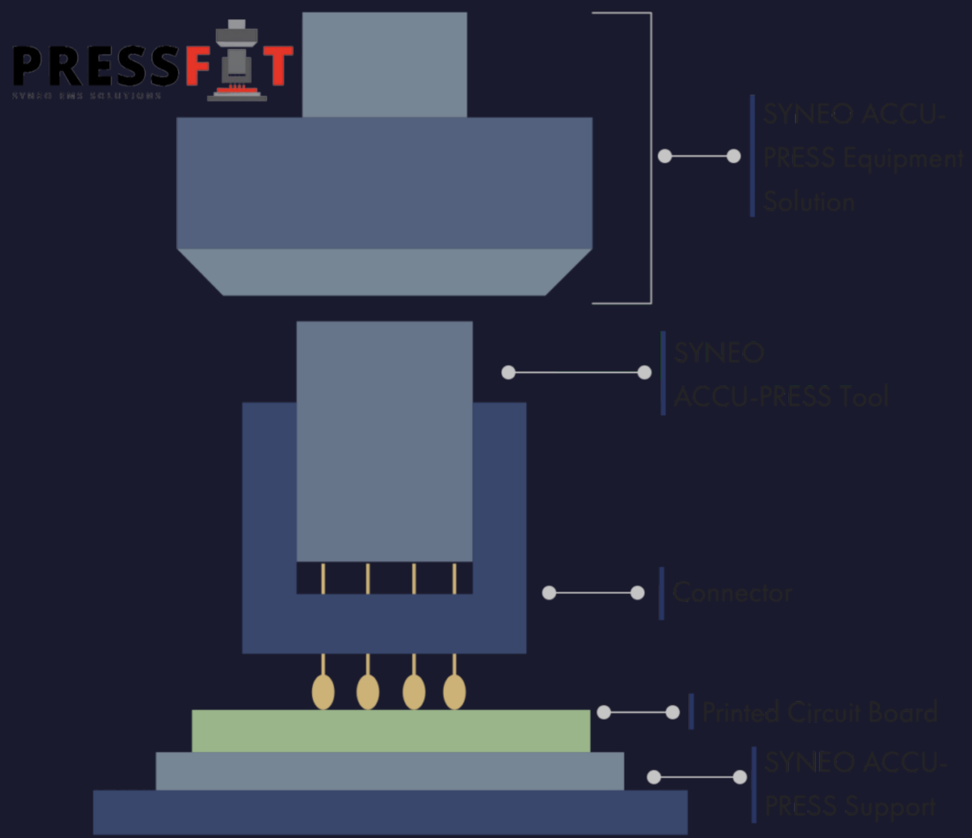


EPT press-fit tool for hm2.0 AB blade strip connector with alignment pins and force distribution ribs

SPIROL Pin Installation Equipment:

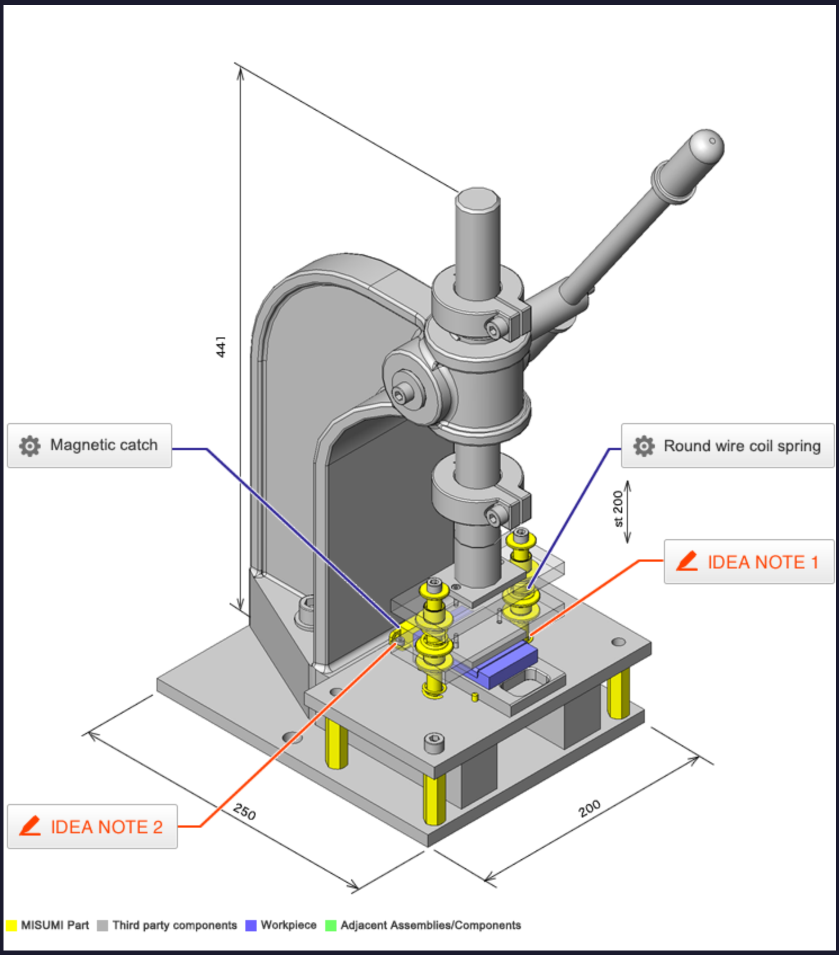
SPIROL pin installation technology showing manual, semi-automatic, and fully automatic press systems

SYNEO Press Connector Equipment Diagram:



SYNEO press-fit equipment diagram showing upper insertion tool, connector, PCB, and lower support fixture

MISUMI Plastic Case Press-Fit Fixture:



MISUMI manual arbor press fixture for plastic case press-fit assembly with magnetic catch and coil spring

9. QUICK REFERENCE — MACHINE & FIXTURE SELECTION

Use this guide to select the appropriate press-fit machine and fixture for your PCBA application:

Application	Machine	Fixture Type	Key Features
Prototype / R&D / Repair	Manual Arbor Press (P10)	Simple aluminum support plate	Low cost, hand-operated, force gauge optional
Low-volume production (<100 units/day)	P50 Bench Machine	Dedicated PCB support fixture + basic top tool	Pneumatic insertion, manual PCB load, force monitoring
Medium-volume (100–500 units/day)	P100 / CSP-3T	Shuttle-mounted fixture + PPS top tool	Auto shuttle, XY positioning, force-distance monitoring
High-volume automotive (500+ units/day)	CSM-5T / CAP-6T	Custom fixture with barcode + PPS tool	Full traceability, vision system, automatic tool change
Inline continuous production	CT-2030-2 / Robotic	SMEMA-compatible fixture + vision alignment	MES integration, 2D code scanning, fully automated
Single-pin repair / sample	P300 / P350	Individual pin support anvil + conversion kit	3.5–5 pins/sec, rotary insertion finger, vision
High-current power elements	Servo press 44–115kN	Heavy-duty steel fixture with large clearance	Force up to 115kN, multiple parallel pins, thermal manage
Backplane / high-speed I/O	CSP with vision + PPS	Precision fixture with datum alignment pins	Fine-pitch alignment, controlled impedance verification

Key Takeaways:

1. Servo-electric presses with force-distance monitoring are mandatory for automotive and high-reliability press-fit assembly.
2. Bottom fixtures must provide support pillars directly under connector sites and clearance holes for pin protrusion.
3. Top tools should press on pin shoulders, not housing, with tolerance ≤3 mils for even force distribution.
4. PPS (Pin Penetration Sensing) tooling provides real-time verification of every pin without destructive testing.
5. Vision systems and barcode/RFID fixture identification prevent operator errors and misalignment.
6. Always sequence assembly: SMT reflow first, then press-fit insertion as a separate mechanical step.
7. Maintain 4mm keep-out zones around press-fit sites to protect nearby SMT components from insertion force transfer.